Physics Domain Hierarchical Lexicons

This document provides CKS lexicons optimized for physics papers.

Registry: [?]

Series Path: [CKS-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-13-2026] → [CKS-MATH-16-2026] → [CKS-DWDM-5-2026] → [CKS-MATH-17-2026] → [CKS-MATH-18-2026] → [CKS-MATH-19-2026] → [CKS-MATH-20-2026] → [CKS-MATH-21-2026]

Parent Framework: [CKS-0-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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OPERATIONAL DECLARATION

This document provides CKS lexicons optimized for physics papers.

Purpose: Enable physics researchers to introduce CKS framework with appropriate depth.

Scope: Only terms relevant to physics applications:

- Particle physics
- Cosmology
- Quantum mechanics replacement
- Constants derivation
- Forces and fields
- Relativity

- Substrate mechanics

Excluded: Biology, consciousness, clinical terms (see separate domain lexicons)

PHYSICS DOMAIN REDUCTION SERIES

Level P0: Complete Physics Set (80 terms)

All physics-relevant terms from CKS framework

Foundational (10 terms)

- **D=3** - Three spatial dimensions (axiom)
- **S=2** - Bilateral symmetry (axiom)
- ** ℚ ** - Rational substrate (axiom)
- **K-Space** - Discrete geometric substrate
- **X-Space** - Rendered continuous space
- **LERP** - Linear interpolation (K→X transform)
- **Discrete** - Countable, not continuous
- **Rational** - ℚ only, no irrationals
- **Hexagonal Lattice** - 2D substrate packing structure
- **Computational** - Reality as geometric computation

Particle/Soliton Physics (15 terms)

- **Soliton** - Stable geometric pattern (particle)
- **Tier** - Hierarchical soliton level (0-6)
- **Tier-0** - Universe scale
- **Tier-1** - Galaxy cluster
- **Tier-2** - Galaxy (2 trillion derived)
- **Tier-3** - Star/stellar object
- **Vortex** - Rotational soliton structure
- **Toroidal** - Donut-shaped geometric pattern
- **String-8** - Figure-8 lemniscate pattern
- **Phase Lock** - Angular synchronization between solitons

- **Registry** - Hierarchical tracking structure
- **N=0 Pivot** - Universal root node
- **J-Distance** - Logarithmic hierarchical depth
- **Mass** - Registry signature (geometric property)
- **Spin** - Intrinsic angular momentum (winding)

Constants & Derivations (20 terms)

- **α_EM = 1/137.036** - Fine structure constant (derived from hexagonal geometry)
- **e = 2.71828** - Euler's constant (manifold saturation)
- ** $\pi \rightarrow 4$** - Pi approaches 4 at relativistic velocities
- **φ = 1.618** - Golden ratio (hexagonal packing)
- **Δ = 19** - Baseline geometric constant
- **R = 69** - Critical threshold (6-9 hook)
- **32** - Base harmonic (2^5)
- **144** - Lepton scaler (12^2)
- **163** - Curvature quantum
- **2.08** - Holographic scale factor
- **7.70164** - Jacobian (7-bubble nucleus)
- **84** - Segmented word length
- **1024** - Bit capacity (2^{10})
- **2 Trillion** - Galaxy count (derived)
- **Zero Free Parameters** - All constants forced by geometry
- **Planck Scale** - Substrate resolution anchor
- **c (speed of light)** - Substrate propagation speed
- **$\hbar$ (reduced Planck)** - Angular momentum quantum
- **G (gravitational)** - Curvature coupling constant
- **Baryon Asymmetry** - 3:1 ratio from boot sequence

Forces & Fields (10 terms)

- **Electromagnetic** - Hexagonal phase coupling
- **Strong Force** - 3-neighbor constraint (6-fold)

- **Weak Force** - Bilateral asymmetry (S=2)
- **Gravity** - Parent soliton opcode (not force)
- **Momentum** - Registry persistence opcode
- **Field** - Substrate density gradient
- **Photon** - Substrate vibration pattern (not particle)
- **Gluon** - 3-way coupling pattern
- **W/Z Boson** - Bilateral exchange patterns
- **Higgs** - Substrate density modifier

Quantum Mechanics Replacement (12 terms)

- **Discrete** - Replaces continuous wave functions
- **Deterministic** - No probability, geometric necessity
- **Non-Local** - Registry chain communication
- **Entanglement** - Shared registry parent
- **Superposition** - Substrate state ambiguity (pre-measurement)
- **Collapse** - Measurement forces discrete state
- **Uncertainty** - Substrate resolution limit (Planck scale)
- **Tunneling** - Phase-space adjacency
- **Wave-Particle** - Single phenomenon (substrate vibration)
- **Observable** - Registry read operation
- **Eigenvalue** - Allowed discrete states
- **Commutator** - Registry operation order

Relativity & Cosmology (8 terms)

- **Spacetime** - K-Space substrate (not curved continuum)
- **Lorentz Contraction** - Hexagonal→square transition
- **Time Dilation** - Registry tick-rate variation
- **Black Hole** - Maximum density soliton
- **Dark Matter** - 3 wing-pairs per hex (18 units)
- **Dark Energy** - Substrate expansion pressure
- **Hubble Constant** - Tier-separation growth rate
- **CMB** - Substrate thermal baseline

Logismos Physics Math (10 terms)

- **Logismos** - Discrete K-Space mathematics
- **VFR Tuple** - (Volume, Frequency, Remainder)
- **Base 32^-1** - Substrate counting base (0.03125)
- **dN** - Discrete differential (not dx)
- **ΣN** - Discrete summation (not ∫dx)
- **Logos (L)** - Pure counting unit
- **dL/dN** - Rate of change in discrete space
- **Substrate Tick** - 1/32 second (base period)
- **Integer Operations** - All calculations in $\mathbb{Q}$
- **Geometric Necessity** - No fitting, forced by axioms

Measurement & Validation (5 terms)

- **Derivation** - From D=3, S=2, $\mathbb{Q}$ to conclusion
- **Measurement Arbitration** - Physical constants validate
- **Falsification** - Specific testable predictions
- **Compilation** - Logical consistency check
- **Zero Tuning** - No free parameters adjusted

Total: 80 terms

Level P1: Standard Physics Paper (50 terms)

Sufficient for most physics publications

Term	Physics Definition
D=3, S=2, $\mathbb{Q}$	Three axioms: 3D space, bilateral symmetry, rational substrate
K-Space	Discrete geometric substrate (computational reality)
X-Space	Continuous LERP-rendered perception space
Hexagonal Lattice	2D substrate structure (honeycomb packing)
Discrete	Substrate is countable, not continuous

Term	Physics Definition
Rational ($\mathbb{Q}$)	Only rational numbers exist (no √2, π is approximated)
Soliton	Stable geometric pattern (replaces “particle”)
Tier (0-6)	Hierarchical nesting (universe, galaxy, star, planet…)
Tier-2 (Galaxy)	2 trillion galaxies derived from tier structure
J-Distance	Logarithmic tier depth ($\log_{32}$ measure)
N=0 Pivot	Universal root node (all registries terminate)
Registry	Hierarchical parent-child tracking chain
Vortex	Rotational soliton structure (all entities)
Mass	Registry signature (not fundamental property)
Spin	Geometric winding number (intrinsic angular momentum)
α_EM = 1/137.036	Fine structure from hexagonal+bilateral geometry
e = 2.71828	Manifold saturation decay constant
** $\pi \to 4$**	Pi approaches 4 at relativistic motion (hex→square)
Δ = 19	Baseline geometric constant
R = 69	Critical closure threshold (6-9 hook)
32 Hz	Base harmonic frequency (2^5)
Zero Free Parameters	All constants geometrically forced
Photon	Substrate cymatic vibration (not particle)
Field	Substrate density/tension gradient
Electromagnetic	Hexagonal phase-coupling mechanism
Strong Force	3-neighbor constraint in hexagonal lattice
Weak Force	Bilateral (S=2) asymmetry mechanism
Gravity	Parent soliton opcode (not attraction force)
Momentum	Registry persistence opcode
Entanglement	Shared registry parent (non-local connection)
Superposition	Pre-measurement substrate ambiguity
Collapse	Measurement forcing discrete registry state
Wave-Particle Duality	Single phenomenon (substrate vibration pattern)
Uncertainty	Substrate Planck-scale resolution limit
Tunneling	Phase-space adjacency crossing
Spacetime	K-Space substrate (not curved manifold)

Term	Physics Definition
Lorentz Contraction	Hexagonal→square geometric transition
Time Dilation	Variable registry tick-rate under motion
Dark Matter	3 wing-pairs per hex unit (18/6 = 3 $\times$ visible)
Dark Energy	Substrate expansion pressure
Black Hole	Maximum density soliton (total closure)
Logismos	Discrete mathematics for K-Space
VFR (Volume, Frequency, Remainder)	K-Space state descriptor
Base 32^-1	Substrate counting base (1/32)
dN (discrete differential)	Replaces dx in K-Space
Derivation	Geometric consequence from axioms
Measurement	Physical constants validate framework
Falsification	Specific tests that would prove wrong
Compilation	Logical consistency verification
Baryon Asymmetry	3:1 matter/antimatter from boot bit-flip

Total: 50 terms

Level P2: Physics Methods Section (30 terms)

Core physics methodology

Term	Essential Physics Definition
D=3, S=2, $\mathbb{Q}$	Framework axioms
K-Space	Discrete substrate (reality)
Hexagonal Lattice	2D substrate structure
Discrete/Rational	Countable $\mathbb{Q}$ -only substrate
Soliton	Stable pattern (particle equivalent)
Tier	Hierarchical level (0=universe, 2=galaxy)
N=0 Pivot	Universal registry root
J-Distance	Tier depth (logarithmic)
Registry	Parent-child tracking
Mass	Registry signature
α_EM = 1/137.036	Fine structure (hexagonal derivation)
e, π , φ	Constants from geometry
Δ=19, R=69	Geometric thresholds
Zero Parameters	All forced by axioms
Photon	Substrate vibration
Field	Substrate gradient
EM, Strong, Weak, Gravity	All geometric mechanisms
Entanglement	Shared registry parent

Term	Essential Physics Definition
Wave-Particle	Single substrate phenomenon
Uncertainty	Planck resolution limit
Spacetime	K-Space substrate
Lorentz	Hex→square transition
Dark Matter	3 wing-pairs (18 units)
Logismos	Discrete K-Space math
VFR Tuple	(V,F,R) state
Base 32^-1	Counting base
dN	Discrete differential
Derivation	From axioms
Measurement	Validates via constants
Falsification	Testable predictions

Total: 30 terms

Level P3: Physics Abstract (20 terms)

Minimal physics context

Term	Abstract-Level Definition
D=3, S=2, $\mathbb{Q}$	Three geometric axioms
K-Space	Discrete substrate
Hexagonal	Substrate geometry
Discrete	Not continuous
Soliton	Stable pattern (particle)
Tier	Hierarchical level
α_EM = 1/137.036	Derived fine structure
Zero Parameters	Geometric necessity
Photon	Substrate vibration
Field	Substrate gradient
Forces	Geometric mechanisms
Entanglement	Shared registry
Wave-Particle	Single phenomenon
Spacetime	K-Space substrate
Dark Matter	Wing-pairs (factor 3)
Logismos	Discrete math
Derivation	From axioms
Measurement	Constant validation
Falsification	Testable predictions
2 Trillion Galaxies	Derived count

Total: 20 terms

Level P4: Physics Ultra-Compact (15 terms)

Minimum for physics comprehension

Term	Ultra-Compact
D=3, S=2, $\mathbb{Q}$	Axioms
K-Space	Discrete substrate
Hexagonal	Substrate structure
Soliton	Particle equivalent
Tier	Hierarchical scale
α_EM = 1/137.036	Derived constant
Zero Parameters	All forced
Photon	Vibration pattern
Field	Substrate gradient
Entanglement	Shared registry
Spacetime	K-Space
Dark Matter	$3 \times$ factor
Logismos	Discrete math
Derivation	From geometry
Measurement	Validates

Total: 15 terms

Level P5: Physics Core (10 terms)

Skeleton understanding

Term	Core Physics
D=3, S=2, $\mathbb{Q}$	Axioms
K-Space	Discrete substrate
Hexagonal	Geometry
Soliton	Particle
α_EM = 1/137.036	Derived
Photon	Vibration
Field	Gradient
Logismos	Discrete math
Derivation	Geometric
Measurement	Validation

Total: 10 terms

Level P6: Physics Absolute Minimum (7 terms)

Cannot reduce further

Term	Irreducible Physics
D=3, S=2, $\mathbb{Q}$	Framework axioms (3D, bilateral, rational)
K-Space	Discrete geometric substrate
Soliton	Stable pattern (all particles/structures)
α_EM = 1/137.036	Fine structure (derived from hexagonal geometry)
Logismos	Discrete substrate mathematics
Derivation	Geometric consequence from axioms
Measurement	Physical constants validate

Total: 7 terms

Level P7: Emergency Physics (5 terms)

Severe constraint only

Term	Emergency
D=3, S=2, $\mathbb{Q}$	Geometric axioms
K-Space	Discrete substrate
α_EM = 1/137.036	Derived constant
Derivation	From geometry
Measurement	Validates

Total: 5 terms

Level P8: Physics Minimum (3 terms)

Single sentence only

Term	Absolute Minimum
D=3, S=2, $\mathbb{Q}$	Axioms
Derivation	Constants from geometry
Measurement	Validates

Total: 3 terms

PHYSICS-SPECIFIC USAGE GUIDE

By Physics Sub-Domain

Particle Physics Papers

Priority additions to base level:

- Soliton, tier, vortex, mass, spin
- Strong force, weak force, electromagnetic
- Photon, gluon, W/Z, Higgs mechanisms
- Entanglement, superposition, collapse
- Baryon asymmetry

Cosmology Papers

Priority additions:

- Tier-0 (universe), Tier-2 (galaxy, 2 trillion)
- Dark matter (wing-pairs), dark energy
- Hubble, CMB, black holes
- Spacetime, expansion
- Lorentz, time dilation

Quantum Mechanics Replacement

Priority additions:

- Discrete, deterministic
- Superposition, collapse, entanglement
- Uncertainty, tunneling
- Wave-particle duality resolution

- Observable, eigenvalue, commutator

Constants Derivation

Priority additions:

- α_EM, e, π , φ
- Δ=19, R=69, 32, 144, 163
- Zero free parameters
- Planck scale
- Baryon asymmetry

Forces & Fields

Priority additions:

- Field as substrate gradient
- EM (hexagonal coupling)
- Strong (3-neighbor)
- Weak (bilateral S=2)
- Gravity (parent opcode)

Common Physics Paper Structures

Theoretical Physics Paper

Abstract: Level P3 (20 terms)

Introduction: Level P2 (30 terms)

Theory: Level P1 (50 terms)

Derivations: Level P0 (80 terms) + equations

Results: Level P2 (30 terms)

Discussion: Level P1 (50 terms)

Brief Letter

Abstract: Level P4 (15 terms)

Body: Level P3 (20 terms)

Conclusions: Level P5 (10 terms)

Conference Abstract

Entire abstract: Level P4 (15 terms)

Poster

Title: Level P8 (3 terms)

Introduction: Level P4 (15 terms)

Methods: Level P3 (20 terms)

Results: Level P4 (15 terms)

Physics Term Introduction Order

Optimal sequence for physics papers:

1. **Axioms** (D=3, S=2, $\mathbb{Q}$) - “Framework rests on three geometric axioms…”
2. **K-Space** - “Reality is discrete substrate, not continuous…”
3. **Hexagonal** - “Substrate has hexagonal 2D packing…”
4. **Soliton** - “Particles are stable geometric patterns…”
5. **Discrete/Rational** - “All values rational ($\mathbb{Q}$), countable…”
6. **Derivation** - “Constants derive from geometry…”
7. **α_EM** - “Fine structure: 1/137.036 from hexagonal bilateral…”
8. **Measurement** - “Validates via physical constants…”
9. Then sub-domain specific terms

Critical Physics Equations

Always include when relevant:

α_EM = $1/(6^2 \times \pi \times S) \approx 1/137.036$

e = lim(1 + 1/N)^N as N $\rightarrow \infty$

$\pi \rightarrow 4$ (relativistic limit)

Dark Matter / Visible = 18/6 = 3

Galaxy Count = f(tier spacing) $\approx 2 \times 10^{12}$

Lorentz $\gamma = (1 - v^2/c^2)$^(-1/2) (hex→square)

Physics-Specific Falsification

Tests specific to physics domain:

1. **Constants**: Do all Standard Model constants derive correctly?
2. **Galaxy count**: Does tier structure predict 2 trillion?
3. **Dark matter ratio**: Is it exactly 3:1 from geometry?
4. **Entanglement**: Does shared registry explain correlations?
5. **Lorentz**: Does hexagonal transition match relativity?
6. **Quantum**: Do discrete states replace wave functions?
7. **Forces**: Do geometric mechanisms match observations?

Comparison to Standard Physics

Key distinctions to emphasize:

Standard Model	CKS Framework
26+ free parameters	0 free parameters
Continuous space	Discrete hexagonal
Quantum probability	Geometric determinism
4 fundamental forces	4 geometric operations
Wave-particle duality	Single substrate phenomenon
Dark matter unknown	3 wing-pairs per hex
Constants unexplained	All derived from D=3, S=2, $\mathbb{Q}$

Recommended Physics Paper Template

Title

Use Level P8 (3 terms): "Deriving [constant/phenomenon] from D=3, S=2, $\mathbb{Q}$"

Abstract (250 words)

Use Level P3 (20 terms):

- State axioms
- Describe discrete substrate
- Present key derivation
- Give measurement match

- State falsification

Introduction (1000 words)

Use Level P2 (30 terms):

- Introduce problem
- Present CKS axioms
- Explain K-Space substrate
- Preview derivation method
- State predictions

Theory (2000 words)

Use Level P1 (50 terms):

- Full axiom explanation
- Hexagonal geometry details
- Soliton mechanics
- Registry structure
- Mathematical framework

Derivation (3000 words)

Use Level P0 (80 terms):

- Step-by-step from axioms
- All relevant constants
- Logismos mathematics
- Geometric necessity shown
- Zero parameters demonstrated

Results (1000 words)

Use Level P2 (30 terms):

- Derived values
- Measurement comparisons
- Agreement quantified
- Discrepancies noted

Discussion (1500 words)

Use Level P1 (50 terms):

- Implications for physics
- Comparison to standard model
- Advantages/disadvantages
- Future predictions
- Falsification tests

Conclusion (500 words)

Use Level P3 (20 terms):

- Summary of derivation
- Measurement validation
- Next steps
- Call for testing

END CKS-LEX-3-2026

Status: Complete Physics Domain Lexicons

Levels: 8 (from 80 terms to 3)

Optimization: Physics papers only

Minimum: 7 terms (Level P6)

For other domains, see:

- CKS-LEX-4-2026: Biology Domain
- CKS-LEX-5-2026: Mathematics Domain
- CKS-LEX-6-2026: Consciousness Domain
- CKS-LEX-7-2026: Engineering Domain
- CKS-LEX-8-2026: Clinical/Medical Domain

Physics lexicon complete and optimized.

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-13-2026**] The Resonant Epoch. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-13-2026

[**CKS-MATH-16-2026**] The Geometric Necessity of 32. Github: https://github.com/ghowland/cks/tree/main/papers/MAT
MATH-16-2026

[**CKS-DWDM-5-2026**] The Geometry of the 66th Harmonic. Github: https://github.com/ghowland/cks/tree/main/papers/I
DWDM-5-2026

[**CKS-MATH-17-2026**] The 7-Bubble Jacobian. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-17-2026

[**CKS-MATH-18-2026**] The 84-Bit Word on a 32-Bit Computer. Github: https://github.com/ghowland/cks/tree/main/paper
MATH-18-2026

[**CKS-MATH-19-2026**] Topological Recirculation. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CK
MATH-19-2026

[**CKS-MATH-20-2026**] The Winding Torus. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-20-2026

[**CKS-MATH-21-2026**] The Final Constant Closure. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
MATH-21-2026